



## Neuroprotective effect of 20(R)-ginsenoside Rg<sub>3</sub> against transient focal cerebral ischemia in rats

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### HIGHLIGHTS

- We investigated the neuroprotective effect of 20(R)-Rg<sub>3</sub> against transient focal cerebral ischemia in rats.
- 20(R)-Rg<sub>3</sub> alleviated neurological deficit and decreased the volume of infarction.
- 20(R)-Rg<sub>3</sub> significantly decreased the number of TUNEL-positive cells.
- 20(R)-Rg<sub>3</sub> reduced the expressions of calpain I and caspase-3 mRNA.
- 20(R)-Rg<sub>3</sub> could protect brain against injury induced by cerebral ischemia in rats.

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### ABSTRACT

Ginsenosides, the active ingredients of Chinese herbal medicine *Panax ginseng*, have a wide spectrum of medical effects, such as anti-tumorigenic, angiostimulatory, adaptogenic, and anti-fatigue activities. In the present study, we have investigated the neuroprotective effect of 20(R)-ginsenoside Rg<sub>3</sub> (20(R)-Rg<sub>3</sub>) against transient focal cerebral ischemia in male Sprague–Dawley (SD) rats. The middle cerebral artery was occluded for 2 h in rats and then reperused for 24 h. The behavioral disturbance was evaluated according to neurological deficit scores, and the infarct volumes were evaluated by 2,3,5-triphenyltetrazolium chloride (TTC) staining; in addition, ischemia-mediated apoptosis was examined using the method of terminal deoxynucleotidyl transferase (TdT)-mediated d-UTP nick end labeling (TUNEL). The expressions of calpain I and caspase-3 mRNA in hippocampal CA1 region were further assayed using in situ hybridization, in order to clarify the neuroprotective mechanism of 20(R)-Rg<sub>3</sub>. 20(R)-Rg<sub>3</sub> at the doses of 10 and 20 mg kg<sup>−1</sup> i.p., but not 5 mg kg<sup>−1</sup>, showed significant neuroprotective effect in rats against focal cerebral ischemic injury by markedly reducing cerebral infarct volumes and degrading infarct rate of TTC-stained coronal brain sections, and improving behavior of the animals. Our results also suggested that 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>−1</sup>) could significantly suppress the expressions of calpain I and caspase-3 mRNA. These results indicated that 20(R)-Rg<sub>3</sub> attenuates the neuronal apoptosis caused by cerebral ischemia–reperfusion injury and its neuroprotective effect may be involved in the downregulation of calpain I and caspase-3.

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Stroke/brain attack resulting from an abrupt blockage of arteries leading to the brain is a kind of serious neurological injury. About 88% of all strokes can be attributed to cerebral ischemia. It forms a major of death and it is an important source of chronic disability all over the world. With regard to the mechanisms of ischemia stroke on the neural injury, the recent studies suggested that it causes not only delayed neuronal death but also apoptosis in dying neurons

after ischemia [14,22]. In spite of major advances of neuroprotective therapeutic approaches for treating ischemic stroke over the last decade, stroke is still a serious problem for which effective drug therapy is not yet available [18]. It is of great concern about new neuroprotective drugs in acute cerebral stroke.

*Panax ginseng*, which has been used for thousands of years, is one of the most widely recognized traditional Chinese medicines. Approximately 40 different types of ginsenosides have been separated and identified from *P. ginseng* [12]. Among the ginsenosides, ginsenoside Rg<sub>3</sub> appears to be responsible for a wide range of therapeutic and pharmacological actions including anti-cancer, anti-inflammation and anti-fatigue *in vitro* and *in vivo* [4,19]. The

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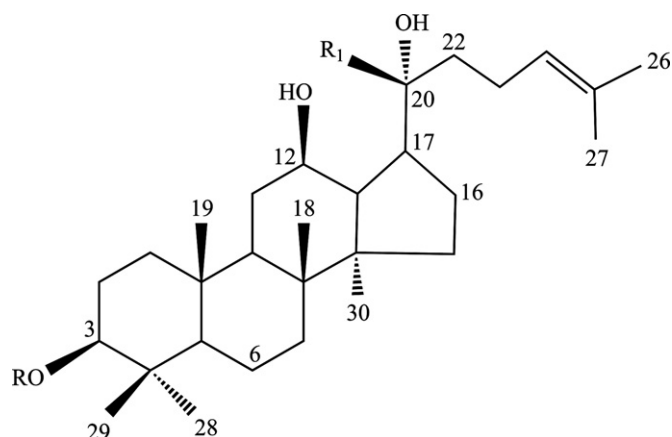


Fig. 1. Chemical structure of 20(R)-ginsenoside Rg<sub>3</sub>.

previous experiments showed that ginsenoside Rg<sub>3</sub> modulated various types of ion channels, such as voltage-dependent and ligand-gated ion channels in neuronal cells, via stimulation of ATP-sensitive K<sup>+</sup> channel, the attenuation of Ca<sup>2+</sup> influx, and inhibition of peak Na<sup>+</sup> channel [7]. In our primary experiments, we found that 20(R)-ginsenoside Rg<sub>3</sub> (20(R)-Rg<sub>3</sub>) could produce the protective effect on lipopolysaccharides-induced cultured human umbilical vein endothelial cell (HUVEC) injury and the mechanism might be associated with suppressing the mobilization of cytosolic calcium, inhibiting the generation of PAI-1, and modulating the balance of tPA/PAI-1 [6]. Recently, oral or intravenous administration of ginsenosides Rg<sub>3</sub> was found to show neuroprotection against the cerebral ischemia-induced injury in rats [20]. The currently ongoing research showed that ginsenoside Rg<sub>3</sub> could protect hippocampal neurons from homocysteine-induced excitotoxicity [9]. Ginsenoside Rg<sub>3</sub> has neuroprotective effect on the prevention of neuronal cell death via the inhibition of NMDA receptor-mediated signals in rat hippocampal neurons and the inhibition of mitochondrial permeability transition pores [10,21]. However, there is no report about the direct evidence of neuroprotective effect of 20(R)-Rg<sub>3</sub> (Fig. 1). It is thus interesting to investigate whether 20(R)-Rg<sub>3</sub> has neuroprotective effect against transient focal cerebral ischemia in rats.

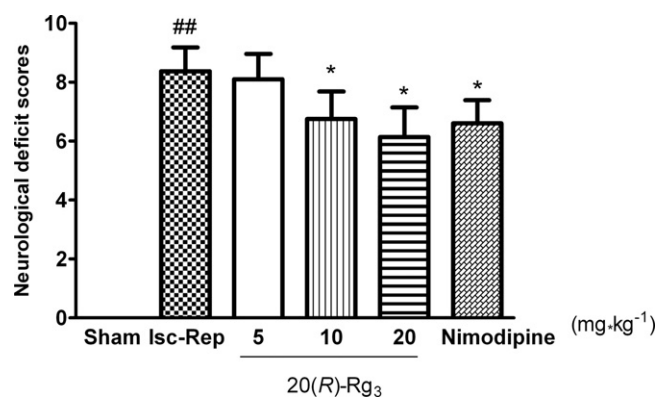


Fig. 2. The neuroprotective effect of 20(R)-Rg<sub>3</sub> on neurological deficit scores 24 h after MCAO in rats. Sham-operated animals did not show any neurological deficit scores. 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>-1</sup>) significantly improved neurological functions. Data represent mean ± SEM, ## *P* < 0.01 versus sham group, \* *P* < 0.05 versus Isc-Rep group. One-way ANOVA followed by Student's *t*-test. *n* = 6.

In this study, we investigated the neuroprotective effect of 20(R)-Rg<sub>3</sub> against transient focal cerebral ischemia in rats. The behavioral disturbance was evaluated according to neurological deficit scores; the infarct volumes were determined by 2,3,5-triphenyltetrazolium chloride (TTC) staining; the apoptosis in situ labeling was detected by terminal deoxynucleotidyl transferase (TdT)-mediated d-UTP nick end labeling (TUNEL). Furthermore, in order to clarify the neuroprotective mechanism of 20(R)-Rg<sub>3</sub>, the expressions of calpain I and caspase-3 mRNA in hippocampal CA1 region were assayed using in situ hybridization.

Healthy, adult male Sprague-Dawley rats weighting between 290 and 350 g were purchased from the preclinical animal house, Kunming Medical University. Animals were housed in standard conditions with maintained humidity (55%), controlled temperature (21–23 °C), and standard 12-h light/dark cycle (light on at 7:00 a.m.). At the same time, food and water were available ad libitum at all times. All the experiments were conducted according to the requirements of NIH Guidelines for care and use of laboratory animals and local legislation on the ethical use of animals. A blinded manner was performed in all animal procedures. In total, 108 rats were randomly divided into the following six groups (*n* = 18 per

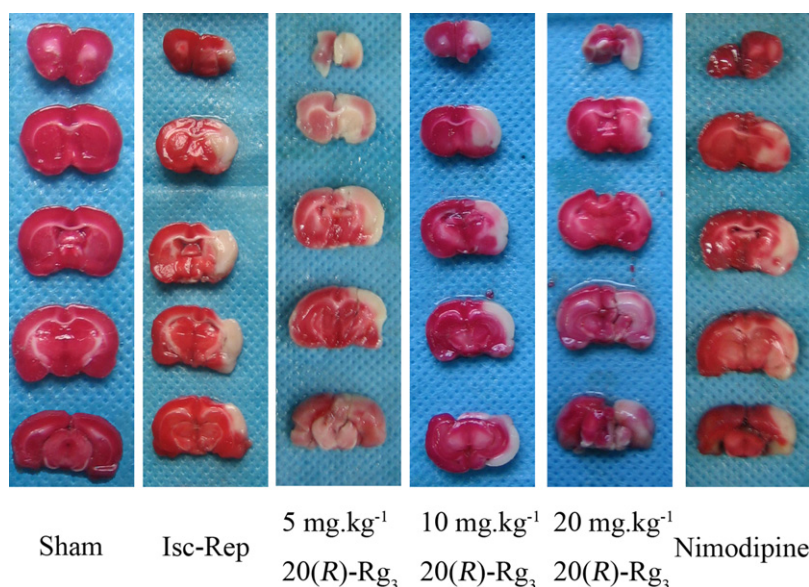


Fig. 3. Neuronal damage was determined 24 h after MCAO in rats by staining with TTC. Quantifications of the infarct volume showed that administration with 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>-1</sup>) significantly decreased infarct formation.

group): (1) sham-operated group (sham); (2) ischemia–reperfusion group (Isc-Rep); (3) 20(R)-Rg<sub>3</sub> low dose group (5 mg kg<sup>-1</sup>); (4) 20(R)-Rg<sub>3</sub> middle dose group (10 mg kg<sup>-1</sup>); (5) 20(R)-Rg<sub>3</sub> high dose group (20 mg kg<sup>-1</sup>); (6) positive control group (nimodipine 1 mg kg<sup>-1</sup>). A total of 36 rats ( $n=6$  per group) were randomly assigned in the assessment of the neurological deficit scores, and another 36 rats ( $n=6$  per group) were randomly selected for evaluation of the infarct volumes. The rest of 36 rats ( $n=6$  per group) were used for TUNEL staining and in situ hybridization staining for calpain I and caspase-3 mRNA expressions. 20(R)-Rg<sub>3</sub> (5, 10 and 20 mg kg<sup>-1</sup>), nimodipine 1 mg kg<sup>-1</sup> and normal saline (for the ischemia–reperfusion group and sham-operated group) were intraperitoneally injected into rats at the volume of 1 ml/100 g body weight, twice daily before occluding the blood supply.

20(R)-Rg<sub>3</sub> (purity >98%) was kindly provided by Prof. Zhihe Chen (Kunming Medical University, PR China). According to the method of the patent (CN1569011-A), it was dissolved in PEG-400. TdT In Situ apoptosis detection kit was obtained from R&D Systems, Inc., USA. Calpain I and caspase-3 mRNA kits were purchased from Boster Bioengineering Company, Wuhan, China. All the other chemicals were of analytical grade and were purchased from Sigma, local commercial suppliers.

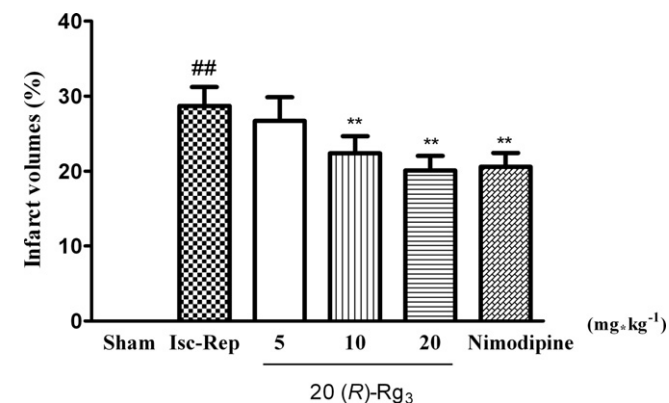
All the surgical procedures were performed by using sterile/aseptic techniques in accordance with institutional guidelines. The animals were subjected to right middle cerebral artery occlusion (MCAO). Rats were anesthetized with a chloral hydrate solution (300 mg kg<sup>-1</sup>, i.p.). MCA occlusion was induced by the intraluminal suture MCAO method as previously described by Longa et al. [11]. After 120 min of focal ischemia, the filament was withdrawn to initiate reperfusion for 24 h.

To evaluate the neuroprotective effect of 20(R)-Rg<sub>3</sub> after transient focal cerebral ischemia, we tested the neurological deficit scores, following the method of Bederson's test [1]. Rats were subjected to activity monitoring for about 2 h for behavioral recording. Accordingly, four categories of neurological finding were noted. Grade 0: no observable neurological deficit, marked 0; Grade 1: failure to extend right forelimb on lifting the whole body by tail, marked 1; Grade 2: circling to the contralateral side, marked 3; Grade 3: leaning to the contralateral side at rest or no spontaneous motor activity, marked 6. The higher the score is, the worse the neurological deficits appear.

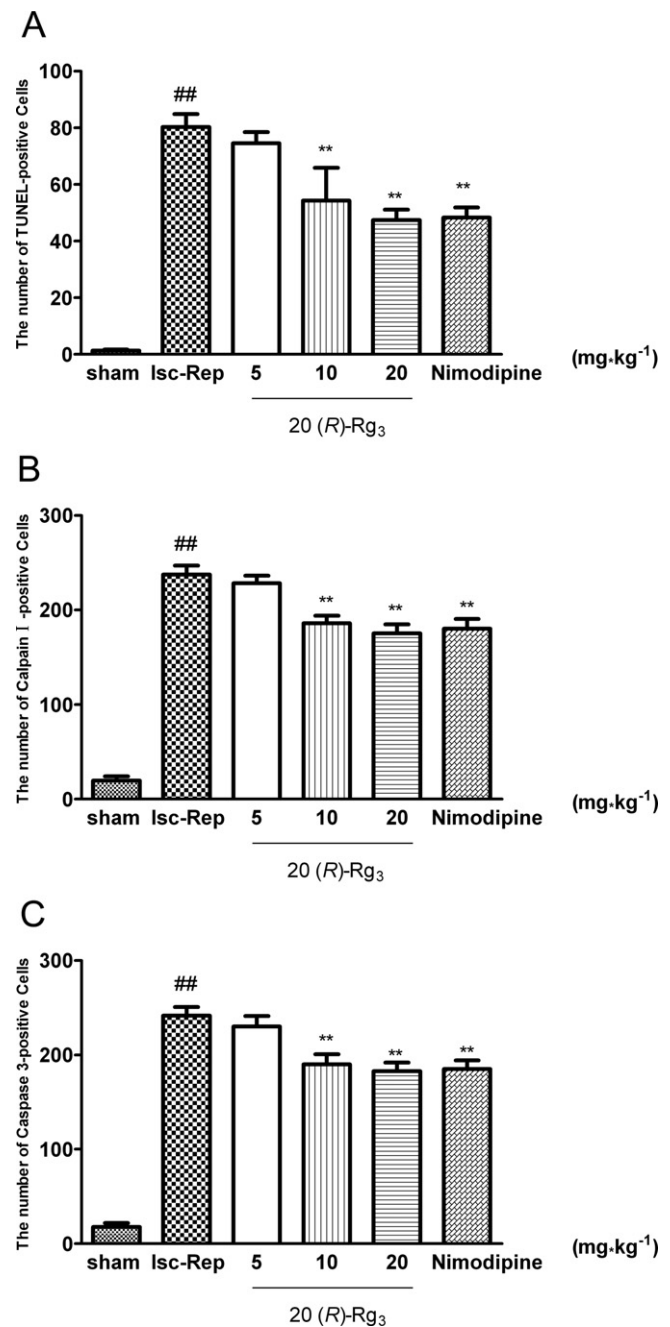
Infarct volumes in the brain were assessed with TTC staining method [8]. Briefly, 24 h after MCAO, the animals were anesthetized and the brain tissues were collected and sliced into 2.0-mm thick

sections. Brain slices were placed in cold phosphate-buffered saline (PBS) for 15 min and incubated in 0.1% TTC solution for 30 min at 37 °C. Then the slices were transferred into 4% formaldehyde solution for fixation. Infarct volumes of the stained brain sections were measured using Image Pro Plus image analysis software. The infarct volume of each brain was calculated by multiplying the infarct area of the slides by its thickness.

Apoptotic cells were detected using TUNEL method. The labeling targets were the multitude of new 3'-OH ends of the DNA fragments and typically localized in morphologically identifiable nuclei

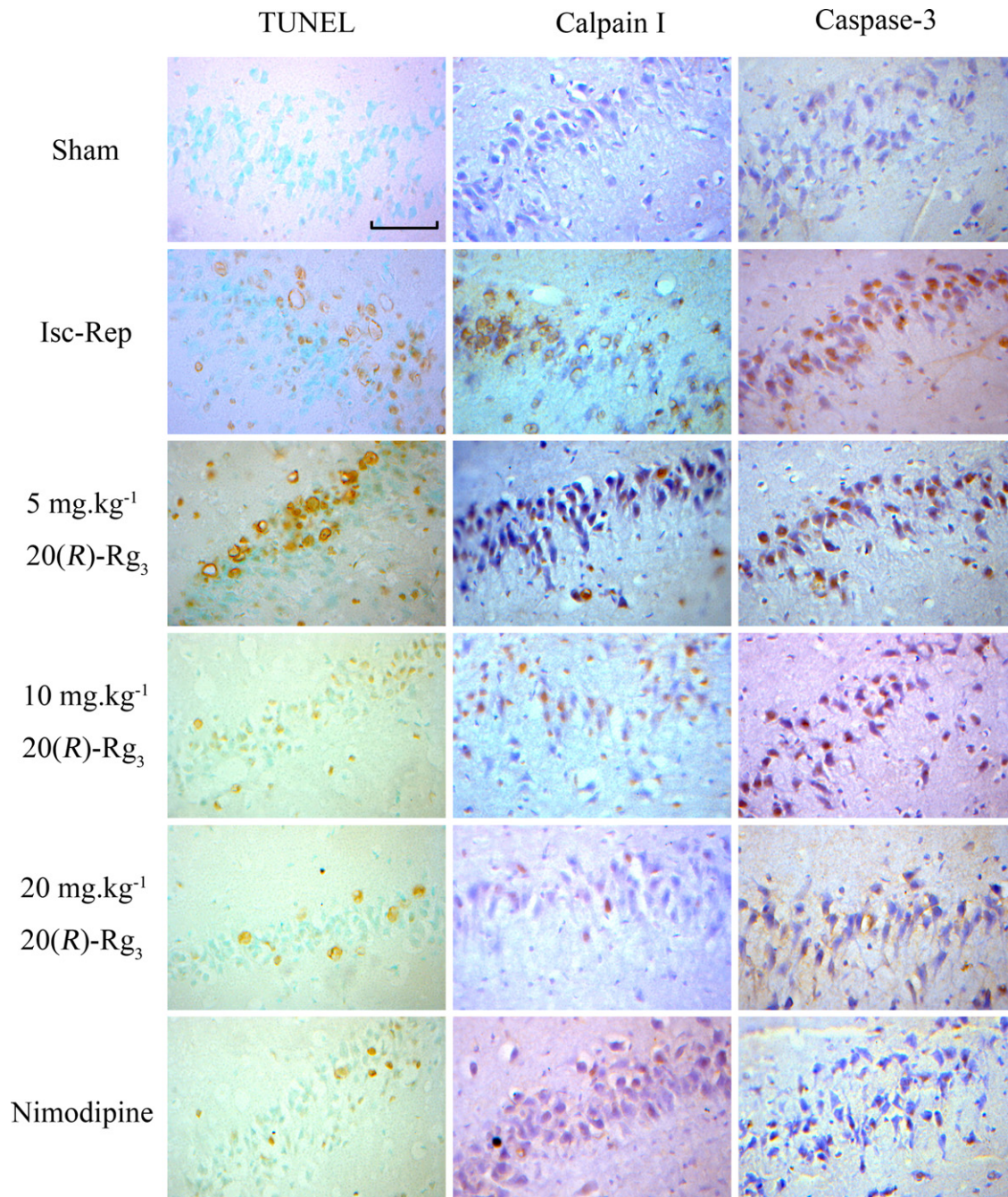


**Fig. 4.** The neuroprotective effect of 20(R)-ginsenoside Rg<sub>3</sub> on infarct volumes 24 h after MCAO in rats. Treatment of 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>-1</sup>) significantly decreased the volume of infarction induced by ischemia injury. Data represent mean  $\pm$  SEM, ##  $P < 0.01$  versus sham group, \*\*  $P < 0.01$  versus Isc-Rep group. One-way ANOVA followed by Student's *t*-test.  $n=6$ .



**Fig. 5.** The neuroprotective effect of 20(R)-ginsenoside Rg<sub>3</sub> on TUNEL assay (A), calpain I (B) and caspase-3 (C) mRNA expressions in hippocampal CA1 region 24 h after MCAO in rats. Treatment of 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>-1</sup>) significantly decreased the number of TUNEL-positive cells ( $P < 0.01$ ), calpain I-positive cells ( $P < 0.01$ ) and caspase-3-positive cells ( $P < 0.01$ ) induced by ischemia injury. Data represent mean  $\pm$  SEM, ##  $P < 0.01$  versus sham group, \*\*  $P < 0.01$  versus Isc-Rep group. One-way ANOVA followed by Student's *t*-test.  $n=6$ .





**Fig. 6.** Representative histological images of TUNEL staining, in situ hybridization staining for calpain I and caspase-3 mRNA expressions in hippocampal CA1 region 24 h after MCAO in rats. In the sham group, few TUNEL-positive cells, calpain I-positive cells and caspase-3-positive cells were present in hippocampal CA1 region. In the Isc-Rep group, the number of TUNEL-positive cells, calpain I- and caspase-3-positive cells was significant increased than that in the sham group. The number of TUNEL-positive cells, calpain I- and caspase-3-positive cells of 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>-1</sup>) significantly decreased compared with Isc-Rep group. The scale bar in left upper panel (100 μm) applies to the all of the pictures.

of apoptotic cells. The detail procedures were carried out according to the instructions of the kits.

Calpain I and caspase-3 levels were evaluated by assaying mRNA expressions. The sequence of probes against calpain I mRNA was: 5'-TCCAT GAGGT TCACC ATGCT GCGGC ACGAC-3'; 5'-GGGCT CCGAG TAGCG GGTGA TGATG AGCTC-3'; 5'-TCAAA GGTCA CAACT CCATC CAGAT CTGTG-3' and caspase-3 mRNA was: 5'-TGGAT AGTCT TTGTT TCAAA ATTAT TAATG GATT-3'; 5'-ATGTG CATGA ATTCC AGCTT GTGCG CGTAC TGT-3'. Hybridization was performed on serial brain tissue slices according to manufacturer's directions.

Data are expressed as mean ± standard error of the mean (SEM). Statistical comparisons were made by analysis of variance (ANOVA) and Student's *t*-test. Differences were considered statistically significant when *P* is less than 0.05.

The neurological scores were 0 in sham-operated group, and 8.4 in model group. Nimodipine 1 mg kg<sup>-1</sup> and 20(R)-Rg<sub>3</sub> at the doses of 10 and 20 mg kg<sup>-1</sup>, but not 5 mg kg<sup>-1</sup>, significantly improved the neurological scores. These data are shown in Fig. 2.

No cerebral infarction in hippocampal CA1 region was observed in the sham-operated group by TTC staining. In contrast, the cerebral infarct volumes in the ischemia-reperfusion group were

significantly higher ( $P < 0.01$ , Fig. 3) than those in the sham-operated group. As shown in Fig. 4, 20(R)-Rg<sub>3</sub> decreased the volume of infarction induced by ischemia injury. The percentage of infarct volumes in 5, 10 and 20 mg kg<sup>-1</sup> 20(R)-Rg<sub>3</sub> groups was (26.7 ± 4.5)%, (22.4 ± 3.2)%, and (20.1 ± 2.8)%, respectively. 20(R)-Rg<sub>3</sub> at the dose of 20 mg kg<sup>-1</sup> showed a similar potency compared with 1 mg kg<sup>-1</sup> nimodipine.

In the sham-operated group, few TUNEL-positive cells per slide were present in hippocampal CA1 region. The number of TUNEL-positive cells in the ischemia–reperfusion group was increased more significantly ( $P < 0.01$ , Fig. 6) than that in the sham-operated group. As presented in Fig. 5A, treatment of 20(R)-Rg<sub>3</sub> at the doses of 10 and 20 mg kg<sup>-1</sup> i.p., but not 5 mg kg<sup>-1</sup>, significantly decreased the number of TUNEL-positive cells.

In the sham-operated group, few calpain I-positive cells were measured in hippocampal CA1 region by in situ hybridization. In the ischemia–reperfusion group, the number of calpain I-positive cells was more than ( $P < 0.01$ , Fig. 6) that of the sham group. At the same time, nimodipine and 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>-1</sup>) could significantly decrease calpain I-positive cells as compared with the ischemia–reperfusion group (Fig. 5B).

In sham group, few caspase-3-positive cells were present in hippocampal CA1 region. Compared with the sham group, the number of caspase-3 positive cells significant increased ( $P < 0.01$ , Fig. 6) in the ischemia–reperfusion group. 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>-1</sup>) significantly decreased the number of caspase-3 positive cells (Fig. 5C).

Following stroke, animals subsequently exhibit a variety of neurological deficits. It is very important to evaluate neurological function outcome after stroke. The Bederson scale is a global neurological assessment that was developed to measure neurological impairments following stroke [5]. Our results revealed that 20(R)-Rg<sub>3</sub> could improve neurological behavior disturbance based on neurological deficit scores.

The most frequently used tool for measuring the efficacy of putative neuroprotective compounds is TTC staining. Brain lesion identified by TTC staining indicates that tissues were irreversibly impaired in mitochondrial function and dehydrogenase activity [2]. Our study demonstrated that treatment with 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>-1</sup>) significantly decreased the volume of infarction following cerebral ischemia stress. It was implied that 20(R)-Rg<sub>3</sub> could improve mitochondrial activity after brain ischemia.

Apoptosis is known as the prototypical form of programmed cell death in neurons during development and adult cell turnover, and it may also occur in a range of neurodegenerative conditions. The morphological characteristics of apoptosis include cell shrinkage, membrane blebbing, karyorrhexis, extensive oligonucleosomal DNA fragmentation, and the formation of apoptotic bodies. TUNEL method has been used to indicate the presence of DNA fragmentation in degenerating neurons after cerebral ischemia by numerous investigators [13]. In our study, the TUNEL-positive cells markedly increased in the ischemia–reperfusion group. 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>-1</sup>) obviously reduced the TUNEL-positive cells. We speculated that reducing neural impairment of 20(R)-Rg<sub>3</sub> might be associated with its attenuating apoptosis.

Calpains, one family of cysteine proteases, are activated by calcium and autolytic processing. Two different types of calpains have been described in the brain, namely I or  $\mu$ -calpain, and II or m-calpain. The main difference for  $\mu$ - and m-calpain is the concentration of calcium required for its activation *in vitro* [15]. Among proteolytic systems involved in the apoptotic machinery, calpains may play a decisive role in the signaling of cell death. Once activation, calpains induce a degradation of cellular substrates such as cytoskeletal proteins, membrane proteins, and phosphatases [17]. Caspases, another family of cysteine proteases with unusual substrate specificity, have been identified as key executioners of

apoptosis. Caspase-3, one member of caspase family, is one of the major “execution phase” proteases acting downstream in the apoptotic cascade [3]. During focal cerebral ischemia, calpain I and caspase-3 are activated and involved in the neuronal death. According to the calpain–calpastatin–caspase hypothesis, the calpain I appears to play an important role in the neuronal death and the caspases are molecular guillotines strategically positioned to behead proteins required for maintenance of cellular homeostasis [16]. The present study demonstrated that 20(R)-Rg<sub>3</sub> could decrease the expressions of calpain I and caspase-3 mRNA in hippocampal CA1 region.

In conclusion, 20(R)-Rg<sub>3</sub> (10 and 20 mg kg<sup>-1</sup>) significantly improved the recovery of neurological deficits, reduced the volume of infarction induced by ischemia injury and decreased the number of TUNEL-positive cells in hippocampal CA1 region compared with the ischemia–reperfusion group, and the neuroprotective mechanisms of 20(R)-Rg<sub>3</sub> may be involved in reducing the expressions of calpain I and caspase-3 mRNA. These results suggest that 20(R)-Rg<sub>3</sub> may be a potential therapeutic application against transient focal cerebral ischemia-induced injury. Our further studies will be focused on whether 20(R)-Rg<sub>3</sub> has neuroprotective effect *in vitro*.

## Conflict of interest

All the authors have no conflict of interests.

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## References

- [1] J.B. Bederson, L.H. Pitts, M. Tsuji, M.C. Nishimura, R.L. Davis, H. Bartkowski, Rat middle cerebral artery occlusion: evaluation of the model and development of a neurological examination, *Stroke* 17 (1986) 472–476.
- [2] A. Benedek, K. Moricz, Z. Juranyi, G. Giger, G. Levay, L.G. Harsing Jr., P. Matyus, G. Szenasi, M. Albert, Use of TTC staining for the evaluation of tissue injury in the early phases of reperfusion after focal cerebral ischemia in rats, *Brain Research* 1116 (2006) 159–165.
- [3] K.M. Boatright, G.S. Salvesen, Mechanisms of caspase activation, *Current Opinion in Cell Biology* 15 (2003) 725–731.
- [4] H. Dong, L.-P. Bai, V.K.W. Wong, H. Zhou, J.-R. Wang, Y. Liu, Z.-H. Jiang, L. Liu, The *in vitro* structure-related anti-cancer activity of ginsenosides and their derivatives, *Molecules* 16 (2011) 10619–10630.
- [5] A. Durukan, T. Tatlisumak, Acute ischemic stroke: overview of major experimental rodent models, pathophysiology, and therapy of focal cerebral ischemia, *Pharmacology Biochemistry and Behavior* 87 (2007) 179–197.
- [6] B. He, P. Chen, L. Yang, X.C. Zhang, F. Ji, Z.Q. Shen, Protective effects of 20(R)-ginsenoside Rg<sub>3</sub> on human umbilical vein endothelial cell injury induced by LPS, *Chinese Pharmaceutical Journal* 44 (2009) 1703–1707.
- [7] S.M. Jeong, J.H. Lee, J.H. Kim, B.H. Lee, I.S. Yoon, J.H. Lee, D.H. Kim, H. Rhim, Y. Kim, S.Y. Nah, Stereospecificity of ginsenoside Rg<sub>3</sub> action on ion channels, *Molecules and Cells* 18 (2004) 383–389.
- [8] C.N. Joshi, S.K. Jain, P.S. Murthy, An optimized triphenyltetrazolium chloride method for identification of cerebral infarcts, *Brain Research. Brain Research Protocols* 13 (2004) 11–17.
- [9] J.H. Kim, S.Y. Cho, J.H. Lee, S.M. Jeong, I.S. Yoon, B.H. Lee, J.H. Lee, M.K. Pyo, S.M. Lee, J.M. Chung, S. Kim, H. Rhim, J.W. Oh, S.Y. Nah, Neuroprotective effects of ginsenoside Rg<sub>3</sub> against homocysteine-induced excitotoxicity in rat hippocampus, *Brain Research* 1136 (2007) 190–199.
- [10] S. Kim, T. Kim, K. Ahn, W.K. Park, S.Y. Nah, H. Rhim, Ginsenoside Rg<sub>3</sub> antagonizes NMDA receptors through a glycine modulatory site in rat cultured hippocampal neurons, *Biochemical and Biophysical Research Communications* 323 (2004) 416–424.
- [11] E.Z. Longa, P.R. Weinstein, S. Carlson, R. Cummins, Reversible middle cerebral artery occlusion without craniectomy in rats, *Stroke* 20 (1989) 84–91.

- [12] J.M. Lu, Q. Yao, C. Chen, Ginseng compounds: an update on their molecular mechanisms and medical applications, *Current Vascular Pharmacology* 7 (2009) 293–302.
- [13] M.M. Martinez, R.D. Reif, D. Pappas, Detection of apoptosis: a review of conventional and novel techniques, *Analytical Methods* 2 (2010) 996–1004.
- [14] P. Nicotera, G. Melino, Regulation of the apoptosis–necrosis switch, *Oncogene* 23 (2004) 2757–2765.
- [15] Y. Ono, H. Sorimachi, Calpains – an elaborate proteolytic system, *Biochimica et Biophysica Acta* 1824 (2012) 224–236.
- [16] A. Rami, Ischemic neuronal death in the rat hippocampus: the calpain–calpastatin–caspase hypothesis, *Neurobiology of Disease* 13 (2003) 75–88.
- [17] A. Rami, R. Agarwal, G. Botez, J. Winckler, mu-Calpain activation, DNA fragmentation, and synergistic effects of caspase and calpain inhibitors in protecting hippocampal neurons from ischemic damage, *Brain Research* 866 (2000) 299–312.
- [18] T. Segura, S. Calleja, J. Jordan, Recommendations and treatment strategies for the management of acute ischemic stroke, *Expert Opinion on Pharmacotherapy* 9 (2008) 1071–1085.
- [19] W. Tang, Y. Zhang, J. Gao, X. Ding, S. Gao, The anti-fatigue effect of 20(R)-ginsenoside Rg<sub>3</sub> in mice by intranasally administration, *Biological and Pharmaceutical Bulletin* 31 (2008) 2024–2027.
- [20] J. Tian, F. Fu, M. Geng, Y. Jiang, J. Yang, W. Jiang, C. Wang, K. Liu, Neuroprotective effect of 20(S)-ginsenoside Rg<sub>3</sub> on cerebral ischemia in rats, *Neuroscience Letters* 374 (2005) 92–97.
- [21] J.W. Tian, S.M. Zhang, G.S. Li, Z.F. Liu, B.M. Xu, 20(S)-ginsenoside Rg<sub>3</sub>, a neuroprotective agent, inhibits mitochondrial permeability transition pores in rat brain, *Phytotherapy Research* 23 (2009) 486–491.
- [22] N. Vanlangenakker, T. Vanden Berghe, D.V. Krysko, N. Festjens, P. Vandenabeele, Molecular mechanisms and pathophysiology of necrotic cell death, *Current Molecular Medicine* 8 (2008) 207–220.